

Can Agile Project Management Be Adopted by Industries Other than Software Development?

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ABSTRACT ■

This research paper presents evidence from an exploratory survey on the use of agile project management (APM) practices and the presence of APM enablers in 19 medium- and large-sized companies from different industry sectors considering innovative projects. The results show that these companies are possibly struggling to use their current management practices in the face of different project challenges. Additionally, the presence of some APM enablers indicates opportunities to adapt the APM theory for different companies other than those in software development. Future research should explore the correlation between APM practices and enablers in order to develop “hybrid” management models for different industries.

KEYWORDS: project management; agile project management; APM practices; APM enablers

INTRODUCTION ■

Project planning and control is a challenge for companies engaged in developing new products and/or technologies. This problem has been discussed by those who have been critical of the use of traditional management methods (e.g., waterfall, linear phased-gate approach, or rigid detailed upfront planning followed by rigid execution) in this type of project (Andersen, 1996; Baccarini, 1996; Dawson & Dawson, 1998; Williams, 1999; Maylor, 2001; Cohn, 2005; Shenhar & Dvir, 2007; Stefens, Martinsuo, & Artto, 2007; Perminova, Gustafsson, & Wikström, 2008). The solution could be in establishing a more “flexible” approach to new product development, adaptable to the contingencies of the project environment (Thomke & Reinertsen, 1998; Verganti, 1999; MacCormack, Verganti, & Iansiti, 2001; Smith, 2007; Biazzo, 2009).

This flexible approach has been recognized by the term agile project management (APM) or “agile methods.” According to the literature, the main focus of APM application has been the software industry (MacCormack, Verganti, & Iansiti, 2001; Qumer & Henderson-Sellers, 2008; Mafakheri, Nasiri, & Mousavi, 2008; Sheffield & Lemétayer, 2013). There is an implicit message in the APM literature reinforcing the need to implement or use APM as a “pure” approach, following the practices, tools, and techniques disseminated in this theory (Highsmith, 2004; Cohn, 2005; Schwaber, 2004). Although there is extensive evidence of effective APM use in the software industry, however, there is a lack of empirical studies in other types of industries and projects.

Considering this scenario, would it be possible to find evidence of APM use in other industrial sectors that do not formally adopt or recognize the use of agile project management theory?

In order to address this question, this research paper presents a conceptual framework to help with the identification of practices and enablers related to the APM approach. The framework was applied in an exploratory survey within a group of Brazilian companies. The investigation covered 19 medium-sized and large-sized companies that are parts of the program of the Center of Reference on Innovation, of the Innovation Center at Fundação Dom Cabral (FDC CRI-Minas, Brazil). A characteristic these companies share is their experience in new product development and the fact that they do not yet formally recognize the use of APM methods. The option to study new product development

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projects is due to similarities with the projects from the software industry, such as creativity and the development characterized by continuous cycles of prototyping and testing.

Challenges to Applying Agile Project Management in New Product Development Projects

The term “agile project management” became known as a result of the dissemination of a set of methods developed specifically for the software industry, including *Scrum* (Schwaber & Beedle, 2001; Schwaber, 2004), *Lean Software Development* (Poppendieck & Poppendieck, 2003), *Crystal* (Cockburn, 2004), *Feature Driven Development* (FDD) (Palmer & Felsing, 2002), *Adaptive Software Development* (Highsmith, 2000), *Dynamic System Development Method* (DSDM) (Stapleton, 1997), and *Extreme Programming* (Beck, 1999). These methods were called “lightweight” or “agile” and their creators joined forces to draw up a manifesto, called the *Manifesto for Agile Software Development* (Beck et al., 2001), which can be considered the first step in the expansion of this approach.

Since then numerous books have been published describing agile project management practices and techniques (Chin, 2004; Highsmith, 2004; Augustine, 2005; Cohn, 2005; Sliger & Broderick, 2008; Wysocki, 2009; Cohen, 2010; Goodpasture, 2010; Cobb, 2011). A broader analysis of these books indicates that most of them are still limited to software development projects. Some authors (e.g., Highsmith, 2004; Chin, 2004) argue, however, that these practices, techniques, and tools can be adapted to other types of products and project environments, whose characteristics resemble software projects that are innovative and have a dynamic development environment experiencing constant change.

This possibility has been investigated and discussed in recent studies (Smith, 2007; Cooper, 2008; Conforto & Amaral, 2010). APM is now widely known as one approach for project

management. Recently, the Project Management Institute (PMI) launched a certification that recognizes practitioners in the use of agile project management methods focused on software development projects, called the PMI Agile Certified Practitioner (PMI-ACP)[®]. Consequently, it is important to clearly define the term “APM.” Hereinafter, the definition of agile project management adopted is “[...] *an approach based on a set of principles, whose goal is to render the process of project management simpler, more flexible and iterative in order to achieve better performance (cost, time and quality), with less management effort and higher levels of innovation and added value for the customer*” (adapted from Amaral, Conforto, Benaissi, & Araújo, 2011; Conforto & Amaral, 2010).

With regard to studies that propose to apply this approach to new product development projects, the main focus is to identify the factors that influence its use. APM theory recommends the use of certain practices and tools, such as the concept of product vision, iterative development; the use of visual artifacts such as boards, panels, and sticky-notes, and so forth (Highsmith, 2004; Chin, 2004; Augustine, 2005; Boehm & Turner, 2004), hereinafter called “practices.” These authors have also recognized some of these “practices” as being dependent on the organization’s environment and the project context in which they are used. As a matter of fact, they consider the existence of some “enablers” necessary for APM implementation. These are factors that may influence the use of practices, techniques, and tools, according to the APM approach. We can cite the use of dedicated and co-located teams and the active customer involvement during the entire product development cycle (Hoda, Noble, & Marshall, 2011; Highsmith, 2004; Chow & Cao, 2008).

Additionally, there is room to explore the necessary “enablers” to having successful implementation and results from the APM application. Some

studies focus on the identification of practices, most of them related to the software industry. Some examples are described in the works of Mafakheri, Nasiri, and Mousavi (2008), Qumer and Henderson-Sellers (2008), and Ganguly, Nilchiani, and Farr (2009). As a rule, these studies have focused on software development companies (information technology, IT) that assume the use of APM practices, tools, and techniques in their projects; in addition, the existence of practices and enablers has not been investigated together. The main source used for the evaluation is the set of principles disseminated in the *Manifesto for Agile Software Development* (Beck et al., 2001), resulting in a low-level description and lack of details.

This paper presents a preliminary research effort to identify practices and enablers together in non-software, but innovative, project environments. It was necessary to elaborate on a conceptual framework to help with the collecting of evidence of the implementation of APM practices, their use, and the presence of APM enablers, regardless of the project type, product, or industrial sector.

A Conceptual Framework to Identify APM Implementation

Several studies (Mafakheri, Nasiri, & Mousavi, 2008; Qumer & Henderson-Sellers, 2008; Ganguly, Nilchiani, & Farr, 2009; Sheffield & Lemétayer, 2013) have provided evidence of the presence of principles promoted by the *Agile Software Development Manifesto* (Beck et al., 2001) in software development projects and possible correlations with management practices and project results. However, the results generated are still inconclusive due to challenges with the empirical investigation of a cause and effect relationship; furthermore, there is consensus that the APM practices and enablers should be investigated together, not separately.

The primary research question addressed here is: Would it be possible to identify the use of APM practices and

#	Search Objectives	Quantity of Articles Retrieved	Quantity of Articles Selected	References in Use
SLR1	To identify studies (case research, surveys, theoretical reviews) that present practices from the APM approach	8,159	59	10
SLR2	To identify studies (case research, surveys, theoretical reviews) exploring enablers related to the APM approach	9,634	133	20

Table 1: Quantity of articles retrieved and selected—Systematic Literature Review.

the presence of enablers in companies recognized to be users of the traditional project management approach from different industry sectors that do not formally adopt or recognize the use of APM theory? By identifying the existence of practices and enablers it would be possible to investigate if the organizational environment is favorable or unfavorable for adoption of the APM approach.

Considering this research question, we decided to build a conceptual framework based on two key elements: practices and enablers. The basic premise of the framework is that the application of the agile management approach is linked to the use of practices, tools, and techniques disseminated in the APM theory; however, its use will depend on the existing “enablers,” which are favorable conditions to the proper application of the APM practices. With a better comprehension of these two dimensions, it will be possible to identify if the necessary conditions are in place for the practices to be implemented, or at the very least, if there is room for their development, regardless of whether these pre-conditions exist or not.

The challenge was to identify the practices and enablers to be observed and compose the conceptual framework; therefore, for this matter, a Systematic Literature Review (SLR) procedure was adopted in a grounded literature analysis in order to help the researchers identify a set of practices and enablers to be empirically observed. The SLR protocol was developed specifically for this work, based on studies from the software development, clinical (medicine), and social research areas (Montori, Swiontkowski, & Cook, 2003; Wright, Brand, Dunn, & Spindler, 2007;

Tseng, Dahm, Poolman, Preminger, Canales, & Montori, 2008; Brereton, Kitchenham, Budgen, Turner, & Khalil, 2007, Levy & Ellis, 2006; Biolchini, Mian, Natali, & Travassos, 2005). The set of articles that composed the SLR was extracted from a list of 87 journals, published over a period of 20 years, from the areas of product development, project management, manufacturing, and enterprise management and it also considered books on APM theory. The main database used in the research and cross-database search was the Web of Science.

The searches were performed first on each journal individually and were organized according to two objectives, as shown in Table 1. The selected articles served as the basis for the identification of practices and enablers related to APM theory (cited in the literature review).

The articles were selected using a series of iterative “reading filters and content analysis,” with the support of a set of inclusion criteria and quality evaluation requirements to attain the article and the objectives of the search as listed in Table 1. For example, SLR1 comprised a set of 8,159 articles. This set was subjected to filter 1, which consisted of reading the title, keywords, and abstract. The articles selected in this step then passed through filter 2, which involved reading the introduction and conclusion.

We then reached the set of articles for complete reading and study, which comprised 192 studies, as per “Quantity of Articles Selected” (see Table 1). Based on the detailed analyses of the articles, new studies were also identified throughout the references (manual cross-search analysis). This final set of selected articles was carefully read in order to identify practices and enablers. The results of the analyses of the SLRs were compiled in a database, and finally, 30 studies were considered for this research. In the following two sections, we explain the elements of the conceptual framework.

APM Practices

A project management practice is a specific type of “management action” that contributes to the execution of a process and that may employ one or more techniques and tools (PMI, 2008, p. 433). The majority of the studies analyzed contained management practices without the detailed descriptions of tools and techniques. The analysis resulted in a set of 23 “actions,” 54 techniques, and 21 tools, which were categorized according to the original source (article or book) as having a tendency toward a “traditional” or “agile” project management approach or, in some cases, both. Table 2 summarizes these results (adapted from Eder, Conforto, Schnetzler, Amaral, & Silva, 2012).

	Actions	Techniques	Tools
Traditional project management approach	9	31	4
Agile project management approach	8	17	1
Cited in both approaches	6	6	16
Total	23	54	21

Table 2: Number of practices identified.

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#	APM Management Practices	References (cited in the literature)
1	Use of the "product vision" concept	Highsmith (2004); Augustine (2005)
2	Use of simple project plan communication tools and processes	Highsmith (2004); Cohn (2005); Chin (2004)
3	Use of iterative planning	Eisenhardt & Tabrizi (1995); Boehm & Turner (2004); Highsmith (2004); Schwaber (2004); Augustine (2005); Cohn (2005)
4	Developing activities using self-managed and self-directed teams in the project plan	Takeuchi & Nonaka (1986); Boehm & Turner (2004); Highsmith (2004); Augustine (2005); Vázquez-Bustello, Avella, & Fernández (2007)
5	Use of self-managed and self-directed teams in the project plan monitoring and updating activities	Takeuchi & Nonaka (1986); Boehm & Turner (2004); Highsmith (2004); Vázquez-Bustello, Avella, & Fernández (2007)
6	Frequently apply project plan monitoring and updating processes	Eisenhardt & Tabrizi (1995); Andersen (1996); Boehm & Turner (2004); Highsmith (2004); Augustine (2005); Cohn (2005)

Table 3: Management practices related to the agile project management approach.

We carried out a comparison among the practices from the traditional literature and the agile management approach literature in order to identify which actions, tools, and techniques were clearly different according to the two approaches. The main purpose was to identify a group of management practices that clearly differentiate the use of the agile project management approach from other approaches. The result was six management practices that are being considered in this study, as indicated in Table 3.

The use of the product vision concept (#1 in Table 3) and simple project plan communication tools and processes (#2 in Table 3) are two practices advocated in the APM theory (Highsmith, 2004; Chin, 2004; Augustine 2005; Cohn, 2005). This differs from the practice of defining the product and project scope (PMI, 2008, p. 112). The concept of "vision" foresees the use of visual tools such as visual boards, sticky-notes, figures, or drawings to provide a simple description of the general design, and also requires a high degree of interactions between team members and active collaboration with customers and key stakeholders. At the same time, it focuses on the core problems and challenges the team will have to face in innovative projects, and helps to identify which key aspects are valued most by the customers or market, not "how" to develop the product. It uses metaphors, figures, and prototypes,

thus differing from the traditional way of presenting the scope of the product design, usually in the form of a work breakdown structure (WBS) or bill of material, with a predominantly textual and detailed description of the project activities and tasks to be performed, complemented by detailed product performance parameters.

Another relevant difference refers to the planning activity and the plan's horizon (#3 in Table 3). Instead of a single plan for the entire project, being revised and improved phase by phase, it is performed iteratively, repeating the planning cycle several times along the project life cycle (Boehm & Turner, 2004; Highsmith, 2004; Schwaber, 2004; Augustine, 2005; Cohn, 2005). This practice, called iterative planning (#3 in Table 3), is in line with one of the fundamental principles of APM theory, in other words, iterative development or the rapid and continuous delivery of parts of the product (Highsmith, 2004; Augustine, 2005; Cohn, 2005), thereby obtaining continuous feedback from the customer to be able to respond to constant changes in requirements, needs, risks, new opportunities, and so forth.

Iterative planning also requires the involvement of the project's team in the planning and controlling of activities. Developing activities using self-managed and self-directed teams (#4 in Table 3) is also relevant to ensure their commitment and involvement in developing the project plan (Boehm &

Turner, 2004; Highsmith, 2004; Augustine, 2005). The team's commitment to and involvement with updating the project plan, help to monitor the progress, and must also be encouraged (#5 in Table 3).

The APM authors explain that involvement in both the planning and control of the project contribute to improve interaction and effective communication among the project's team members (Boehm & Turner, 2004; Highsmith, 2004; Augustine, 2005), as well as the development of professionals who can learn, operate, and adapt in the face of complex project environments (Winter, Smith, Morris, & Cicmil, 2006). This practice has also been discussed in the area of product development since the 1980s (Takeuchi & Nonaka, 1986) and also in manufacturing (Vázquez-Bustello, Avella, & Fernández, 2007). Moreover, it is one of the principles described in the *Agile Manifesto* (Beck et al., 2001).

In iterative planning, the activities of the project plan are updated more frequently (#6 in Table 3). According to APM theorists (Boehm & Turner, 2004; Highsmith, 2004; Augustine, 2005; Cohn, 2005), the plan should be updated at the end of each iteration (e.g., short development cycles, on average up to 4 weeks). Within the perspective of the traditional project management theory, in general, the project plan should be revised after a major milestone or a phase conclusion, although this is not considered a rule.

Organization	Process	Project Team	Project Type and Others
1. Organizational structure type 2. Organizational culture 3. Entrepreneurial culture 4. Learning organization 5. Agile-style work environment 6. Acceptance of agile methodology 7. Adequate reward for agile use 8. Emphasis on speed 9. Performance measuring 10. Knowledge management systems 11. Multidisciplinary teams 12. Resource competition 13. Strong executive support 14. Decentralized decision making	1. Capability of reconfiguration 2. Process automatization 3. Process modularity 4. Easy access to information 5. Formalization 6. Frequent development milestones 7. Process concurrency 8. External integration	1. Self-direct teams 2. Team autonomy to make decisions 3. Team leadership 4. Team dedication 5. Team knowledge about agile 6. Team experience/expertise 7. Project manager experience 8. Team size 9. Team location 10. Multidisciplinary team	1. Product succession planning 2. Urgency to complete the project (pace) 3. Goal clarity 4. Project complexity 5. Project newness 6. Support systems, computer-aided design (CAD); computer-aided engineering (CAE) 7. Customer involvement 8. Collaborative work 9. Suppliers' involvement

Table 4: Enablers from the systematic literature review (Adapted from Almeida et al., 2012).

However, unlike the APM approach, this review focuses solely on correcting deviations, because the project plan has already been detailed up through the end of the project.

It should be noted that the practices described in this study do not represent the majority of practices, tools, and techniques comprised in APM theory, but only a small and selected sample, focusing on the main critical differences that characterize the use of the APM approach. These practices were selected based on an extensive investigation of the literature (Eder et al., 2012), by means of a systematic literature review as described earlier.

APM Enablers

In the APM literature it is not difficult to find discussions targeting the importance of the pre-conditions or existing factors that are needed for the successful use of APM practices, tools, and techniques. Some theorists often list principles that are, in fact, conditions to be observed and developed. In this study we refer to such conditions as “APM enablers.” The term, “enablers,” was borrowed from texts in the area of manufacturing, which discuss the concept of “agility” in organizations and manufacturing processes (Vázquez-Bustelo, Avella, & Fernández, 2007). In this research paper, “APM enablers” are defined as “internal or external factors to

the organization that are directly or indirectly related with the implementation of the agile project management approach that may impact the performance and use of a given practice, technique or tool” (adapted from Almeida, Conforto, Silva, & Amaral, 2012). Table 4 illustrates a set of potential APM enablers extracted from the literature.

Forty-one enablers were identified and classified into four categories: organization, process, project team, and project type, as shown in Table 4 (adapted from Almeida et al., 2012). The ones that appeared most frequently and are cited in the articles extracted from the theory are investigated in this study, as indicated in Table 5. The focus of this study is not to describe in detail each enabler, because they are widely discussed in areas such as product development, manufacturing, organization theory, and project management. Additionally, this work is limited to the empirical observation and characterization of these enablers in a group of 19 companies.

Field Research Method

The conceptual framework presented in the previous section was deployed into a questionnaire with multiple-choice questions, based on the tables of selected practices and enablers (see Tables 3 and 5). The questionnaire was divided into three sections:

(1) Characterization of the company, respondent, and types of product development projects; (2) Identification of APM practices; and (3) Identification of APM enablers. Because the objective of this study was an exploratory survey, the questionnaire was created without considering a single standard scale. Each question allowed only one response alternative that best described the reality of the company regarding the practices and the enablers identified in the literature.

The questionnaire was applied at a group of companies that participate in the community of practice organized by the Fundação Dom Cabral of Minas Gerais (MG, Brazil), called the Reference Center on Innovation of Minas Gerais (FDC CRI-Minas). The Dom Cabral Foundation is one of the main institutions for executive education programs and benchmarking for professionals and business managers of large Brazilian corporations.

The companies that participate in the FDC CRI-Minas meet regularly to exchange experiences and knowledge about different innovation-related topics. In one of these meetings, representatives from the 19 companies in the state of Minas Gerais discussed the topic, “how to manage innovative projects.” The questionnaire was emailed before the meeting to professionals involved in the areas of project management,

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#	APM Enablers	References (cited in the literature)
1	Organizational structure type	Vázquez-Bustelo, Avella, & Fernández (2007); Chen, Damanpour, & Reilly (2010); Liu & Yetton (2007); Chow & Cao (2008); Shenhar (2001)
2	Multidisciplinary project teams (various competences)	Eisenhardt & Tabrizi (1995); Vázquez-Bustelo, Avella, & Fernández (2007); Verganti (1999); Zirger & Hartley (1996); Olausson & Berggren (2010)
3	New product development process formalization level	Chin (2004); Highsmith (2004); Li & Atuahene-Gima (1999); Chen, Damanpour, & Reilly (2010)
4	Customer/stakeholder involvement in the product development process	Boehm & Turner (2004); Highsmith (2004); Augustine (2005); Stockstrom & Herstatt (2008); Hoda, Noble, & Marshall (2011); Chow & Cao (2008)
5	Supplier or partner involvement in the product development process	Eisenhardt & Tabrizi (1995); Zirger & Hartley (1996); Chen, Damanpour, & Reilly (2010)
6	Project team members' experience (in years)	Yusuf, Sarhadi, & Gunasekaran (1999); Highsmith (2004); Chow & Cao (2008); Chen, Damanpour, & Reilly (2010)
7	Project manager experience (in years)	Eisenhardt & Tabrizi (1995); Yusuf, Sarhadi, & Gunasekaran (1999); Highsmith (2004); Chow & Cao (2008); Chen, Damanpour, & Reilly (2010)
8	Project team size (number of project team members)	Smith & Reinertsen (1992); Boehm & Turner (2004); Highsmith (2004); Chow & Cao (2008); Qumer & Henderson-Sellers (2008); Mafakheri, Nasiri, & Mousavi (2008)
9	Project team dedication (time dedicated exclusively for the project)	Highsmith (2004); Cohn (2005); Vázquez-Bustelo, Avella, & Fernández (2007); Chow & Cao (2008); Zirger & Hartley (1996); Chen, Damanpour, & Reilly (2010); Smith & Reinertsen (1992)
10	Project team location	Verganti (1999); Zirger & Hartley (1996); Chin (2004); Highsmith (2004); Augustine (2005); Chow & Cao (2008); Qumer & Henderson-Sellers (2008); Chen, Damanpour, & Reilly (2010)

Table 5: Selected enablers related to the agile project management approach.

R&D, and product development. All the companies answered and returned the questionnaires, totaling 48 respondents. The unit of analysis was the project. The respondents were asked to consider one project with some degree of innovation to answer the survey. The companies in question are distributed in the following industrial sectors: mining (5), steel and metallurgical industry (4), auto industry (3), energy (2), engineering projects (2), consumer goods (1), electronics (1), and telecommunications (1), for a total of 19 companies.

As for the sizes of the 19 companies in terms of revenue (in U.S. dollars), five reported revenues of over US\$5 billion; eight reported between US\$0.5 and US\$5 billion; three made between US\$50 and US\$500 million, and three reported revenues of up to US\$50 million. All the participants in this study have been working directly in the areas of projects and/or innovation, and are distributed as follows: 55% are managers of product development or R&D departments, 39% are analysts, and 6% are directors.

Because APM is a theory at an intermediate stage of development (Edmondson & McManus, 2007), the scales and variables are new and need to be further explored; therefore, it was decided to use techniques of qualitative analysis, content analysis, exploratory statistics, and preliminary tests (Edmondson & McManus, 2007). Each question or element of the questionnaire (related to practices and enablers) was analyzed individually.

The primary purpose of this research effort was to identify the presence of APM practices and enablers in companies operating in different industry sectors, in addition to the software industry. These analyses contributed to deepening our empirical knowledge about this research topic in these companies. The results underpinned the discussion of the theoretical implications for this community of researchers and professionals. The data are presented according to the group of 19 companies (or, as per the unit of analysis, 19 projects), using as a reference the percentage of companies

for each question or item observed. A simple mean and median were applied to identify the most latent responses of companies that had more than one respondent per project in the same company.

Results and Analysis

The Use of APM Management Practices

Table 6 summarizes the percentage of companies that presented APM-related practices. For example, in practice 3 (Table 6), 58% of the companies develop the plan "in macro form at the beginning and in detail in each phase." Regarding practice 6, 42% check the progress performance status on a monthly basis (see Table 6). These findings corroborate with the predominant form of planning in these types of companies (e.g., detailing by phase or throughout the development of the project). This means that using a traditional approach was not a surprise in this set of companies because of their nature, mainly from the manufacturing area, which usually adopts traditional project management methods.

Practice (number as per Table 3/Questionnaire)		Scale/Results re Shown in Percentages		
#	Aspect Evaluated	Traditional Approach	Intermediate	Agile Approach
3	Project planning approach	Detailed then revised 37%	Macro then detailed in each phase 58%	Developed by iteration 5%
6	Project plan updating frequency	End of each phase 37%	Monthly 42%	Weekly 21%
1	Project scope description (detailing)	Purely textual and detailed 10%	Textual description with additional information 37%	Minimal textual description (then detailed and revised) 53%
2	Tool used to communicate the project plan	Use of project scheduling tools, such as Gantt charts or WBS 68%	Use of text editor or spreadsheets 32%	Use of visual panels and boards, pictures, drawings, and so forth —
4	Project planning responsibility	Created by a department or the project management office (PMO) 5%	Created by the project manager 48%	Created collaboratively; shared responsibility 47%
5	Project's plan progress and updating responsibility	Department or PMO responsibility 16%	Project manager responsibility 68%	Shared responsibility 16%

Table 6: Management practices observed in the sample.

In this regard, just one company (5%) reported the use of iterative planning (developed by iteration), in other words, developed in cycles. This could be a rich example of a company operating in traditional sectors, generally applying traditional management models but using the iterative planning for some projects with some degree of innovation. Nevertheless, the most interesting evidence was the predominance of the use of phased planning instead of detailing the entire plan at once, then following with execution. It is relevant because of the fact that somehow, some of these companies have challenges creating a complete and detailed project plan. The two pieces of evidence make the hypothesis that one alternative is to use iterative planning for the projects with technology and innovative challenges rather than creating a detailed plan upfront.

An analysis of practices 1 and 2 has provided evidence that the companies tend to employ traditional practices, with tools such as Gantt charts and text editors or spreadsheets, and textual documents to describe the project's scope

and communicate the project plan to the team and stakeholders. The novelty of these findings, however, lies within the high percentage of respondents in the citing of practice 1, "minimal textual description" (53%) which is a favorable indication of the agile project management approach or adaptation of current practices. It is important to consider that the companies did not mention visual boards, pictures, drawings, and so forth (practice 2) as disseminated in the APM approach. We assumed that, in general, these companies are constrained by the traditional mindset and continue to use written and very textual scope statement documents. They most likely faced barriers doing this in innovative projects and have created a simpler way to generate these documents. If they had been familiar with the "vision concept," perhaps they could have adapted this concept to create better project scope documents in their specific project environments.

With regard to practice 4 (see Table 6) "project planning responsibility," 48% of the companies declared that this is the project manager's

responsibility; however, 47% of the companies have declared that the project plan is somehow "created collaboratively, with shared responsibility" (e.g., the project plan is developed in collaboration with the project team members). This is a practice indicated in the APM approach (Boehm & Turner, 2004; Highsmith, 2004; Augustine, 2005). As for the responsibility for updating the project plan (practice 5, Table 6), the most traditional form of data collection predominates (68% of companies), in which the project manager is responsible for collecting and updating the project progress. For 16% of the companies, this activity is totally under the responsibility of a specific department, which in this case, could be the engineering department responsible for the project, or the project management office (PMO), which could be responsible for the administrative activities of the project (PMI, 2008, p.11). For the other 16% of the companies, this is a shared responsibility, which is more suitable for the APM approach.

Project team self-management and shared responsibility are discussed in

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the APM theory as fundamental management practices (Boehm & Turner, 2004; Highsmith, 2004; Augustine, 2005). The emphasis on simplicity, iterative development and customer involvement in the product development process requires that the team members have a strong grasp of the technical aspects of the product, as well as of the managerial aspects. For years, self-management has also been discussed in the classical theory of product development (Takeuchi & Nonaka, 1986) and in texts from the manufacturing sector related to the “agile manufacturing” concept (Vázquez-Bustelo, Avella, & Fernández, 2007). According to the APM approach, it is important, for example, for each team member to be able to identify and understand the possible risks and impacts of a change in the product or

project, and its influence on the results of the project, the company’s goals, and the value it provides for the customer in order to act properly.

In short, among the practices examined here, we found three practices that companies are adopting with adherence to the APM theory. For example, the frequency of updating the project plan (practice 6, Table 6), the use of a minimal textual description, rather than a detailed description and revised to define the project scope (practice 1, Table 6), and the shared responsibility to create the project plan (practice 4, Table 6). The researchers did not expect this result because these companies declare themselves as not being users of the agile management approach. Furthermore, there are companies applying some variance of iterative planning

(practice 1, Table 6); the predominance of the phased planning approach and the application of a less textual description of the project scope, combined with minimum revisions of plans and the high percentage of team members involved in the development of the project plan. This result can be interpreted as relevant evidence pointing out that some companies (deliberately or not) are moving toward the use of agile management combined with traditional management practices.

Thus, the emerging hypothesis is that these companies, as well as software development companies are confronted with the limitations of traditional project management approaches and are incorporating “flexible” management approaches (e.g., agile management practices) at least in the projects aimed

**Enablers (as per Table 5/
Questionnaire)**

		Scale					
#	Question	Traditional Approach		Intermediate		Agile Approach	
6	Project team members’ experience	No previous experience —	<1 year —	1–2 years 21%		2–3 years 79%	>4 —
7	Project manager experience	No previous experience 10%	<1 year 11%	1–2 years 11%		2–3 years 26%	>4 42%
8	Project team size (number of people)	More than 50 5%	21–50 5%	13–20 6%	9–12 21%	6–8 21%	1–5 42%
10	Project team members’ locations	Virtual teams 10%		Same site 21%	Same building 58%	Same room 11%	
4	Customer/stakeholder involvement in the project planning	Not involved 5%		Involved without influence 58%		Involved with influence 37%	
5	Supplier and partner involvement	Not involved 10%		Development phase 53%	Conception phase 21%	All project phases 16%	
1	Organizational structure type	Weak matrix 16%		Balanced matrix 37%	Strong matrix 31%	Project oriented 16%	
3	Product development process formalization		Standardized 37%			Partially standardized 63%	
2	Multidisciplinary project teams	Not multi-functional 21%		Some key departments 68%		Most of the departments 11%	
9	Project team dedication (time allocated to work on the project)	<25% of time 32%	26%–50% of time 31%	51%–75% of time 16%		76%–90% of time 16%	Full time (>90%) 5%

Table 7: Enablers observed in the sample.

at developing highly innovative products under dynamic environments that require a more responsive, fast-learning-execution approach.

The Presence of APM Enablers

Table 7 describes the companies in terms of the presence of enablers or organizational conditions that may be ideal or useful for APM implementation. In general, the project team has over two years of experience (79%), and the experience of the project manager can also be considered relevant (i.e., over two years of experience in 68% of the companies) as described in the Enabler 6 and Enabler 7 columns, respectively (Table 7). According to Table 7, this looks like the best fit for APM approach implementation and it is aligned with the APM theory; however, this doesn't mean that a lack of experience is irrelevant in the traditional approach.

The companies have small project teams (up to 12 professionals in 84% of the cases), as described as the "project team size" (Enabler 8, Table 7). There is also a tendency to employ multidisciplinary project teams (68%) grouping competences from key departments, even though they do not cover all areas of the organization (Enabler 2, Table 7).

All these enablers contribute to the APM implementation. The major challenge is regarding the co-location (Enabler 10, Table 7) and team dedication (Enabler 9, Table 7). Only 11% of the companies had teams working in the same location (e.g., in the same room), and only 5% had team members working full time on their projects. Most projects have teams working on projects on a part-time basis only (e.g., from 26% to 50%, or less than 25% of the total time available allocated). We can imply that these projects are coordinated and executed along with other projects, simultaneously.

Team "size" and "location" are aspects that have been studied since the 1990s in the product development research (Smith & Reinertsen, 1992; Zirger & Hartley, 1996; Verganti, 1999).

These authors provided evidence on innovative projects regarding the more necessary use of small teams, co-located and being fully allocated. Agile management-related texts (Boehm & Turner, 2004; Highsmith, 2004; Augustine, 2005), focusing on software development projects, describe the explicit need for working with small co-located teams (up to 15 professionals). In addition, dedicating time and effort to a single project is strongly recommended by some APM authors (Highsmith, 2004; Cohn, 2005), especially for those with a high degree of innovation. This allows for greater concentration on the work, focus and commitment, creativity and interaction among team members, supported by a shared and holistic vision of the project's challenges and goals.

Hence, albeit only intuitively, some companies of the sample already organize their innovative project teams in accordance with this trend, which reinforces the importance of these enablers for innovative business environments. On the other hand, for some types of projects and companies, the most important element is the easy access to data to enable faster decision making. This is supported by Carbonell and Rodriguez's (2006) argument, which highlights that more important than the team's co-location is the rapid and easy access to information and team members.

These data reveal an important aspect for future research. Are communication tools and use of virtual teams (as 10%, according to Enabler 10, Table 7) capable of delivering the same result of co-located teams in agile project management projects? The evidence collected suggests that researchers who intend to investigate agile practices for the non-software industry should consider the development of management models, practices, and tools aiming to consider distributed or virtual team members, as it has previously been discussed in the software development area (Ramesh, Cao, Mohan, & Xu, 2006; Hossain, Ali Babar, & Paik, 2009; Lee & Yong, 2010).

The active involvement of customers/stakeholders plays a fundamental role in the APM approach. The literature indicates several positive aspects, including increased speed of innovation by the firm; it avoids excessive costs of features that are not demanded by customers; reduces the time and cost of product development; reduces errors related to business requirements; there is a higher product quality through better understanding of customer needs; and a greater ability to negotiate customers' expectations (Benassi, Junior, & Amaral, 2011). Additionally, the supplier involvement in product development can also help to reduce cycle time and avoid delays, and in principle, may affect the application of APM (Filippini, Salmaso, & Tassarolo, 2004). Collaborative development (co-design) with suppliers and partners presents challenges, but has become a common practice in industry (Trygg, 1993; Filippini, Salmaso, & Tassarolo, 2004).

The survey indicates that half of the companies (58%) have involved the customer/stakeholders in the project planning but without any influence (Enabler 4, Table 7). One reasonable explanation could be the challenge to involve customers due to unavailability (e.g., have easy and direct access to them), and another reason is that the majority of clients participated in the project development didn't have *the responsibility and accountability* as per the "product owner," as defined in the APM approach. However, 37% of the companies are with early customer/stakeholder involvement, which is consistent with the APM theory. Given the type of company that manufactures consumer goods, there is an inherent difficulty in including the customers due to the geographical distance or challenges in reaching them, and the challenges involved in prototyping products. Therefore, this result would not be expected in this sample of industries. Furthermore, the suppliers' involvement occurs late in the process; in general, they are involved during the development phase after the pre-design

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or concept phase (Enabler 5, Table 7). These characteristics do not favor the implementation of the agile management approach, because the feedback possible from customers and suppliers is required to be given as soon as possible after the early stages of the project.

With regard to the “organizational structure type” (PMI, 2008, p. 28), Enabler 1 (Table 7), was identified mainly as a balanced matrix (37%) or strong matrix structure (31% of the companies), which could be useful for APM implementation, although not ideal. Agile management literature does not mention which is the best organizational arrangement for this approach. Despite this, the authors (Highsmith, 2004; Cohn, 2005; Chow & Cao, 2008; Chen, Damanpour, & Reilly, 2010) claim the need for full dedication from the project team, which means that having project-oriented organizational structures would be the best, because they are more likely to have fully dedicated project teams. The project-oriented and strong matrix structure should be the best condition for implementation of the APM approach. They could contribute by providing total autonomy in decision making for the project manager and the project team; almost half of the studied companies have project-oriented or strong matrix structures as shown in Table 7.

The enabler “product development process formalization” (Enabler 3, Table 7) is considered the positive aspect for manufacturing firms working with complex and innovative projects. The studied companies have experience in the use of project management models and practices; therefore, they presented evidence of institutionalized product development processes, so we expected to find 100% of the projects following well established and formalized processes and practices. Nevertheless, the majority of projects (63%) were considered as using “partially” standardized processes. It is unexpected and interesting evidence. It is a sign of the pitfalls and limitations these companies have

been faced to properly align their product development processes to fit innovative project needs. This result shows the same tendency observed in software developing companies: the search for new management models for innovative and fast changing dynamic projects, which has resulted in the rise of the agile project management approach.

It should be noted that the partially formalized process could contribute toward a better adaptation to different types of projects, with a higher or lower degree of innovation. This also would allow the application of “agile approaches,” which are more suitable for each type of project to be incorporated, assumed, and defended by APM authors such as Boehm and Turner (2004). These researchers emphasize the need for a sufficient balance between agile practices and formalization to allow for reliability, responsiveness, traceability, and quality of results. Therefore, in order to properly apply APM in more innovative companies it would be ideal to identify the balance between formalization and flexibility, pursuing the development of leaner processes.

In summary, there is evidence pointing out that the companies surveyed, aside from their industry sector, are facing similar issues and challenges as those of software companies in the development of innovative products. The companies surveyed, in most cases, are performing innovative projects using experienced and cross-functional teams managed by experienced project managers. Additionally, there is a tendency to have less formalized processes, and therefore empowering the team with some degree of autonomy to make decisions. Moreover, it was clear that the existence of challenges can be negative for APM implementation, such as: difficulties in having co-located team members; the need for functional managers due to the type of organizational structure; and the difficulties in having full-time dedicated project team members working on one project at a time.

Final Remarks and Future Research

This study presents an exploratory survey within 19 medium-sized and large-sized companies operating in different industrial sectors, with respect to practices and enablers related to the implementation of the agile management approach to new product development projects.

The results indicate evidence of favorable aspects for APM implementation; for example, the favorable presence of some APM enablers, such as: the project team and project manager experience; the project team size; and the new product development process formalization level. These aspects indicate that the companies surveyed have some characteristics and organizational enablers similar to companies from the software industry, which is considered a source of motivation to develop and pursue the application of agile management practices. Additionally, there is evidence of APM practices being applied in these companies, including: creating the project scope with minimal textual description (practice 1, Table 6); creating the project plan collaboratively with shared responsibility (practice 4, Table 6); and using a plan updating approach on a weekly basis (practice 6, Table 6).

These companies are experiencing the use of some agile management practices even though they do not belong to the software industry sector. Deliberately or not, project management practitioners are trying to adapt APM practices for their contexts and challenges. This is probable, as these companies might be struggling to adapt their current and formalized project management processes to fit a more innovative portfolio of projects.

Furthermore, this study identified the potential barriers for APM implementation in more “traditional industries.” There are some enablers that can challenge the use of the agile project management approach, such as: the restriction to assign full-time dedication

project teams; the challenge of co-locating all project team members; the difficulty in creating large multidisciplinary teams (with all project competences involved); the challenge of involving customers with a high degree of influence in project development; and the most superficial involvement of suppliers.

Thus, the final answer to the title of this article is considered positive. We judge that this exploratory study has met sufficient results to propose the hypothesis that the APM approach could be adapted to non-software companies, or more traditional industry sectors, at least for innovative projects or even for some parts of the project that require a more flexible management approach. The project management research community should further investigate how to develop “hybrid” management models, considering APM and traditional approaches, in order to balance the “agility” needs and barriers identified.

Unlike software development companies, manufacturing organizations and companies operating in similar sectors as those studied here, are usually more complex in terms of the quantity and interaction between parties; the level of complexity of the products; the product’s cost; the technological challenge and high cost to prototype the product in the early development stages; the number of professionals involved in the design phase, and ultimately, team members who are often dispersed globally. Compared with software development projects, their projects usually have a long-term development cycle. If these characteristics are considered as restrictions in similar companies, then the APM theory should evolve in order to overcome these obstacles and be recognized as one possible approach for some specific types of projects in this kind of particular environment.

One key aspect for the evolution of the APM as an alternative management approach for the types of companies studied here is the investigation on how to work with distributed project

team members; in other words, how to “scale” APM. The use of APM in globally distributed teams is still evolving and there is a lack of compelling empirical results. According to our results, this is one critical limitation that emerged from the analysis that should not be overestimated due to the complexity of working with distributed team members, even considering the traditional project management approach. In this sense, APM should be further explored and adapted for these circumstances and contingencies.

Another way to develop “hybrid” management models is to have a deep understanding about the role of the “agility” construct in the project management theory as well as its relation with management practices and project performance. This assumption raises new questions: Which APM practices are more relevant for the “agility”? Do “hybrid” management models contribute to “agility” and product and project results? In addition, future work should also focus on finding evidence regarding the correlation between enablers and practices, which is represented by the question: What APM enablers are more relevant to supporting APM implementation? It is important to investigate whether or not APM enablers contribute to the successful implementation of APM practices.

Last, this study does not address or discuss the results or effects of the implementation of APM practices and its correlation to the APM enablers, or even, project and product performance. So, there is room to investigate some statements of causality by raising the following questions: Does the use of APM practices and enablers contribute to better product and project performance? Does the combination of practices and enablers result in better “agility” in the project management perspective? And, furthermore, what does “agility” really mean for the project management theory and the deployment of “hybrid” management models?

Like all types of exploratory research, this study raises more questions than

answers and presents limitations. First, the research faced the challenge of defining all questions that composed the survey with standardized questions and scales. Nevertheless, we didn’t find well-developed instruments, as a result of the newness of the APM theory and the lack of studies in the project management literature. Second, it does not allow for generalizations. The results and evidence collected are relevant only for the sample of companies investigated in this study on a broader perspective. The questionnaire was applied to a few professionals in the organizations investigated and the answers were related to a single project. Thus, the findings encourage further exploration with a larger set of companies and multiple projects, including multiple research methods; for example, in-depth case studies to collect more detailed data regarding the use of management practices and the presence of enablers.

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